

Case study

Prediction of concrete compressive strength using support vector machine regression and non-destructive testing

Wanmao Zhang¹, Dunwen Liu^{*}, Kunpeng Cao¹

School of Resources and Safety Engineering, Central South University, Changsha 410083, China



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ABSTRACT

Performance assessment of existing building structures, especially concrete compressive strength assessment, is a crucial aspect of engineering construction for most industrialized countries. Non-destructive testing (NDT) techniques are commonly employed to assess the compressive strength of concrete structures. However, existing methods for predicting concrete compressive strength using NDT techniques and machine learning methods do not take into account the concrete mix proportion design. This study proposes an effective method to predict concrete compressive strength by combining NDT tests with different mix proportion designs and curing ages. Specifically, support vector machine regression (SVR) and back propagation neural network (BPNN) models are established. Furthermore, various machine learning evaluation indexes are utilized to assess the model performance. To construct and validate the prediction models, a total of 180 datasets containing concrete specimens with different mix proportion designs and curing ages are collected from the existing research literature. The prediction results show that the coefficients of determination (R^2) of the SVR and the BPNN prediction models for the test set of concrete compressive strength are 86.0 % and 86.7 % without considering the concrete mix proportion design. The R^2 of the prediction results of the SVR model is higher than 95 % when considering the effects of concrete mix proportion design and curing age. The R^2 of the BPNN prediction model ranged between 92 % and 97 %. All the evaluation indexes of the SVR model for predicting the compressive strength of concrete are better than those of the BPNN model. Consequently, the SVR model can be utilized to accurately evaluate concrete compressive strength during the structural performance assessment of existing buildings.

1. Introduction

With the increasing development of high-rise buildings and infrastructure construction both domestically and internationally, the repair and renovation of existing building structures have become crucial research areas in current engineering construction [1]. Among various building materials, concrete, widely utilized in construction, relies on its compressive strength as a primary indicator of structural quality [2,3]. Generally, the compressive strength of concrete is evaluated mainly by destructive testing method [3–5]. However, destructive tests not only incur high costs but also compromise the performance of the existing structure, making them less feasible for deteriorating concrete structures [4,6]. Therefore, non-destructive testing (NDT) techniques have gained popularity for

^{*} Corresponding author.E-mail address: 235501005@csu.edu.cn (D. Liu).¹ Contributing authors.

assessing the quality of concrete structures, particularly in terms of compressive strength [5,7–9]. Compared to destructive testing, NDT reduces the testing workload, minimizes damage to the original structure, and decreases testing costs, while still providing reliable and accurate results [10]. To enhance the accuracy of predicted outcomes, many studies recommend employing combinations of NDT methods to estimate concrete compressive strength [11,12].

More specifically, non-destructive techniques were proposed by many scholars in the middle of the last century to evaluate the compressive strength of concrete. Among these techniques, the ultrasonic pulse velocity (UPV) and rebound hammer number (RN) tests are the most widely adopted for evaluating compressive strength [12,13]. Furthermore, NDT provides comprehensive information on concrete properties and homogeneity assessment [14–16]. Consequently, various NDT techniques can be employed to evaluate concrete compressive strength. Previous studies have discussed the UPV and RN methods either independently or in combination. Typically, combining UPV and RN measurements yields a more accurate estimation of concrete strength. Some literature proposes merging these two NDT methods to create a precise prediction model, while others utilize a single NDT method for strength prediction [12,17]. Some studies have concluded that the effectiveness of the combined method, compared to individual methods, depends on the number of specimens used in the calibration process [12,13].

The preliminary application of NDT in concrete is mainly based on the classification of concrete according to its mechanical properties. One of the most representative is the classification of concrete quality proposed by Whitehurst. He proposed the characterization of concrete quality with ultrasonic velocity, which can be classified into five quality classes [18]. Other researchers also proposed to characterize concrete quality based on the RN values of the concrete under study. In 1949, Leslie and Cheesman [19] proposed a more specialized concrete classification using ultrasonic velocity measurements and RN values to classify concrete quality into four categories.

In the 1960s, a large number of empirical equations for estimating the compressive strength of concrete based on UPV and RN values were proposed by many researchers and subjected to multiple regression analysis [20,21]. These equations were determined by regression analysis on a series of experimental datasets, including power functions, bilinear functions, and biexponential functions [22–24]. In addition, Lawson et al. [25] investigated the relationship between UPV and the compressive strength of concrete. The relationship between UPV and the compressive strength of concrete increases with age, but the rate of increase is dependent on the mix proportion designs and curing method. Sbartaï et al. [26] proposed a method for obtaining UPV and compressive strength data for concrete using a comprehensive database of NDT experiments. They also conducted an evaluation of existing NDT techniques for assessing concrete compressive strength. Jain et al. [27] combined different NDTs to estimate the compressive strength of concrete. Ali-Benyahia et al. [2] proposed some models between destructive testing and individual or combined NDTs for a more accurate assessment of concrete strength. The results of the study showed that the combined method is of high value in concrete strength estimation compared to the existing models. Ali Poorababi et al. [1] carried out nondestructive testing of a large number of concrete specimens for UPV and RN. An effective Response Surface Method (RSM) was proposed for estimating the compressive strength of concrete, which is more accurate than other available models. Khuram Rashid et al. [8] tested the UPV, RN, and compressive strength of concrete at different curing ages, respectively. The relationship between UPV-compressive strength and RN-compressive strength was established. Their combination was used to predict compressive strength and the correctness of the predicted values was verified. These equations were primarily used to estimate the compressive strength of concrete using UPV values, RN values, or NDT combination measurements. However, there is a high bias in the estimation of these relationships. The estimated compressive strength values differed from the actual values directly measured through destructive laboratory testing.

Traditional models focus on mathematical analysis methods, which mainly include statistical analysis based on linear or nonlinear regression equations. Due to the nonlinear relationship between concrete composition and compressive strength, it is difficult to obtain accurate regression equations, which requires a great deal of skill and experience. Machine learning techniques are important tools for the prediction of nonlinear relationships and have a wide range of applications in civil engineering [28–31]. Some of the machine learning techniques used to develop prediction models for complex problems include Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Relevance Vector Machines (RVM) [32–34]. Among them, artificial neural networks are most widely used to solve the problem of prediction of concrete compressive strength by machine learning techniques [35–38]. Bilgehan and Turgut [39] proposed an ANN model based on the UPV results of specimens to predict concrete compressive strength. The developed model showed a high degree of accuracy and the model can be used to evaluate the compressive strength of concrete. Erda et al. [40] used artificial neural networks and multiple regression techniques to predict the compressive strength of vacuum-treated concrete. The performance of the proposed single and multiple regression models in predicting the compressive strength of concrete was compared. The performance of the developed artificial neural network prediction model using NDT results was superior to the regression equations. Atici [41] also used multiple regression techniques and artificial neural network to predict the compressive strength of concrete. The results showed that the artificial neural network model had better prediction performance than multiple regression. Lorenzi's study showed that artificial neural network prediction can obtain concrete compressive strength at a lower computational cost. Emad et al. [42,43] developed four different soft computing techniques to predict the compressive strength of ultra-high performance fiber concrete and found that the ANN model was more suitable than the others. Mahmood et al. [44] found that the compressive strength of cemented grouted sand can be well predicted using the ANN model. Alkayem et al. [45] generalized and summarized the research on predicting the performance of concrete at high temperatures using machine learning and deep learning. Skentou et al. [46] combined ultrasonic pulse velocity, Schmidt hammer rebound number and effective porosity NDT metrics, and ANN modeling to predict granite unconfined compressive strength. Asteris and Mokos [47] developed multiple back-propagation neural network models. They proposed three different models to predict the compressive strength, including a single input with only UPV values or RN values, and predicting the output using both UPV and RN value numbers. In all cases, satisfactory results were obtained during the training and testing phases. Asteris et al. [5] established several soft computing techniques to estimate concrete compressive strength using UPV and rebound

hammer tests. The BP neural network based on UPV and rebound values is proposed to be the most accurate predictor of concrete compressive strength.

Since the existing methods of predicting concrete compressive strength using NDT and machine learning methods fail to account for the concrete mix proportion design and curing age, both of which can significantly impact concrete strength development [48]. To address this gap, data sets comprising 180 concrete strength specimens with varying mix proportion designs and curing ages were collected from the literature to establish an effective and reliable method for predicting compressive strength [3]. This study proposes a novel approach that combines concrete mix proportioning and NDT (specifically, UPV and RN) to predict compressive strength, leveraging machine learning computational techniques to improve the accuracy and reliability of predictions. Support vector machines and neural networks are widely used in engineering applications due to their proven predictive ability, making them ideal for this study [49,50]. Accordingly, SVR and BPNN are used to predict compressive strength based on NDT data, with consideration given to concrete mix proportion design and curing age. The resulting SVR and BPNN models are evaluated using various machine learning performance metrics to assess their effectiveness in predicting compressive strength.

2. Research significance

Concrete, as the most commonly used building material in construction and infrastructure development, plays a crucial role in determining the structural performance of buildings. Among various indicators, the assessment of compressive strength stands out as a key parameter for evaluating the quality and durability of concrete structures. Despite numerous research efforts focusing on NDT techniques to assess and forecast concrete compressive strength, there is a growing interest in integrating NDT methods with machine learning approaches to enhance prediction accuracy in construction engineering. Existing literature highlights a gap in the current methodologies for predicting concrete compressive strength, as they often overlook key factors such as concrete mix proportion design and curing age. These factors play a significant role in influencing concrete strength development, ultimately impacting the performance of regression models employed for compressive strength prediction.

In this study, we address this gap by collecting a comprehensive dataset comprising 180 concrete strength specimens with diverse proportion designs and curing ages sourced from literature. By integrating concrete mix proportion design with NDT test results, we aim to develop a robust and accurate predictive model for concrete compressive strength. Notably, our research introduces the novel approach of utilizing SVR and BPNN models to predict the compressive strength of concrete specimens, leveraging information from both concrete mix proportion design and NDT tests (UPV and RN). Furthermore, we conduct a thorough evaluation of the proposed models using various machine learning performance metrics to assess their efficacy in predicting concrete compressive strength.

3. Source of dataset

The first step in predicting the compressive strength of concrete using NDT and machine learning methods is to create a sample dataset. This section focuses on the NDT testing methods and dataset sources for the sample data.

3.1. NDT methods

The UPV testing is a widely used NDT method that measures the velocity of a compression stress wave (p-wave) in a material [12]. The principle of the UPV test is based on the propagation velocity of an ultrasonic pulse through the material. The equipment used for the test typically consists of a transmitter, a receiver, and a display device that shows the time taken for the pulse to travel through the material. The pulse velocity is commonly defined as the distance traveled by the pulse divided by the time taken [24].

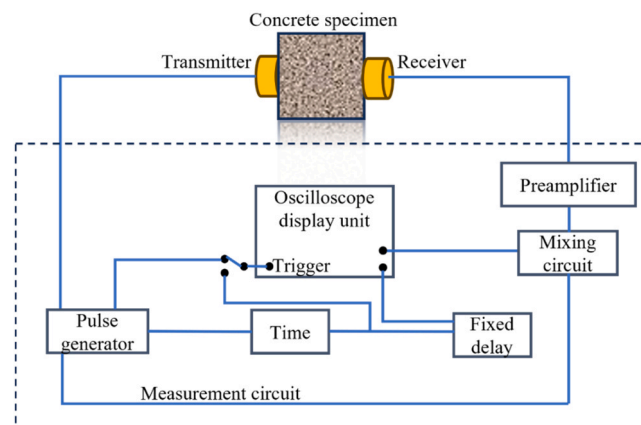


Fig. 1. Schematic diagram of UPV test.

$$V = \frac{L}{T} \quad (1)$$

where V denotes the pulse velocity (m/s); L denotes the path length (m); and T is the transfer time (s).

The UPV method is widely employed in NDT for construction materials. It can be conducted either in a laboratory setting or at the construction site. This method is applicable for various purposes, including strength estimation, uniformity evaluation, durability analysis, surface crack depth measurement, and determination of the dynamic modulus of elasticity for materials such as concrete, rock, and wood [51–55].

The UPV method relies solely on the elastic properties of the material and is independent of its geometry. It is an effective approach for assessing the quality of concrete [56]. The principle of the UPV test is illustrated in Fig. 1. This test is used to evaluate the strength, uniformity, and quality of concrete. A higher pulse velocity indicates good quality and uniformity of the concrete, while a lower pulse velocity suggests the presence of non-uniformities such as cracks or voids. UPV test methods can be categorized as direct, semi-direct, or indirect tests.

The rebound hammer test is a widely used NDT method for assessing the elastic properties and strength of concrete. It operates on the principle of rebound, where the energy absorbed by the concrete surface is related to the strength of the specimen [13,57]. Proper test preparation and selection of the specimen testing surface are crucial for obtaining accurate results. The test surface should be carefully prepared by smoothing it with an abrasive stone. A certain amount of energy is then applied to the surface by striking it with a hammer. The plunger of the hammer must be perpendicular to the smooth surface, and the rebound numbers are recorded, with at least 10 records taken for each surface tested.

Due to the challenge of establishing a direct correlation between the RN and the concrete strength, the rebound hammer test is most suitable for large areas with similar types of concrete. It is recommended to combine this test with other well-known non-destructive tests to obtain a comprehensive assessment of concrete strength. Numerous studies highlight the advantages of the rebound hammer test for concrete, while also emphasizing the need for complementary tests to accurately determine concrete strength.

3.2. Dataset

Based on the existing literature, the experimental design program was developed by incorporating the work of Ali Poorarbabi and Mohammadreza Ghasemi et al. [1,4]. The experimental designs consisted of 20 different concrete mix schemes and the concrete mix proportion design is shown in Table 1. The UPV, RN, and compressive strength of the concrete were measured and recorded at 7 days (7d), 28 days (28d), and 90 days (90d). Consequently, a total of 180 specimen test datasets were obtained. The statistical properties of the relevant sample data are presented in Table 2.

The frequency distribution of the UPV, RN, and concrete strength data was analyzed, and the results are presented in Fig. 2. To assess the relationship between the model input and model output sample data, Pearson correlation coefficients were calculated and are shown in Table 3. Furthermore, a heat map of the Pearson correlation coefficients for the corresponding sample data is depicted in Fig. 3.

Table 3 and Fig. 3 present the Pearson correlation coefficients calculated by comparing the UPV, RN, and concrete compressive strength sample data. The variable with the highest linear Pearson correlation coefficient with compressive strength is RN, followed by UPV. There is some linear correlation between RN, UPV, and concrete compressive strength. While traditional multiple regression analysis can establish correlation equations to predict concrete compressive strength, the prediction results often exhibit high bias.

Table 1
Concrete mix proportion design.

No.	Water-cement ratio	Cement (kg/m ³)	Coarse gravel (kg/m ³)	Fine gravel (kg/m ³)	Sand (kg/m ³)
1	0.79	273	0	936	863
2	0.79	244	497	478	936
3	0.74	291	0	927	980
4	0.74	261	493	473	928
5	0.69	313	0	915	968
6	0.69	280	487	469	919
7	0.65	332	0	920	957
8	0.65	297	529	509	925
9	0.61	354	0	909	949
10	0.61	316	524	504	921
11	0.58	372	0	905	945
12	0.58	332	530	490	908
13	0.54	400	0	890	888
14	0.54	357	528	488	860
15	0.5	432	0	935	872
16	0.5	386	550	469	855
17	0.45	480	0	908	825
18	0.45	429	533	412	800
19	0.42	514	0	884	753
20	0.42	460	558	419	768

Table 2
Statistical properties of model input and output parameters.

Variant	Model inputs							Model outputs		
	Water-cement ratio	Cement (kg/m ³)	Coarse gravel (kg/m ³)	Fine gravel (kg/m ³)	Sand (kg/m ³)	UPV (km/s)	RN	7d compressive strength (MPa)	28d compressive strength (MPa)	90d compressive strength (MPa)
Maximum value	0.79	514	558	936	980	4.9	36	31.2	38.8	38.7
Minimum value	0.42	244	0	412	753	3.37	14	11.3	15.9	17.1
Average value	0.60	356.2	261.5	692	891	4.23	23.1	20.1	24.6	26.7

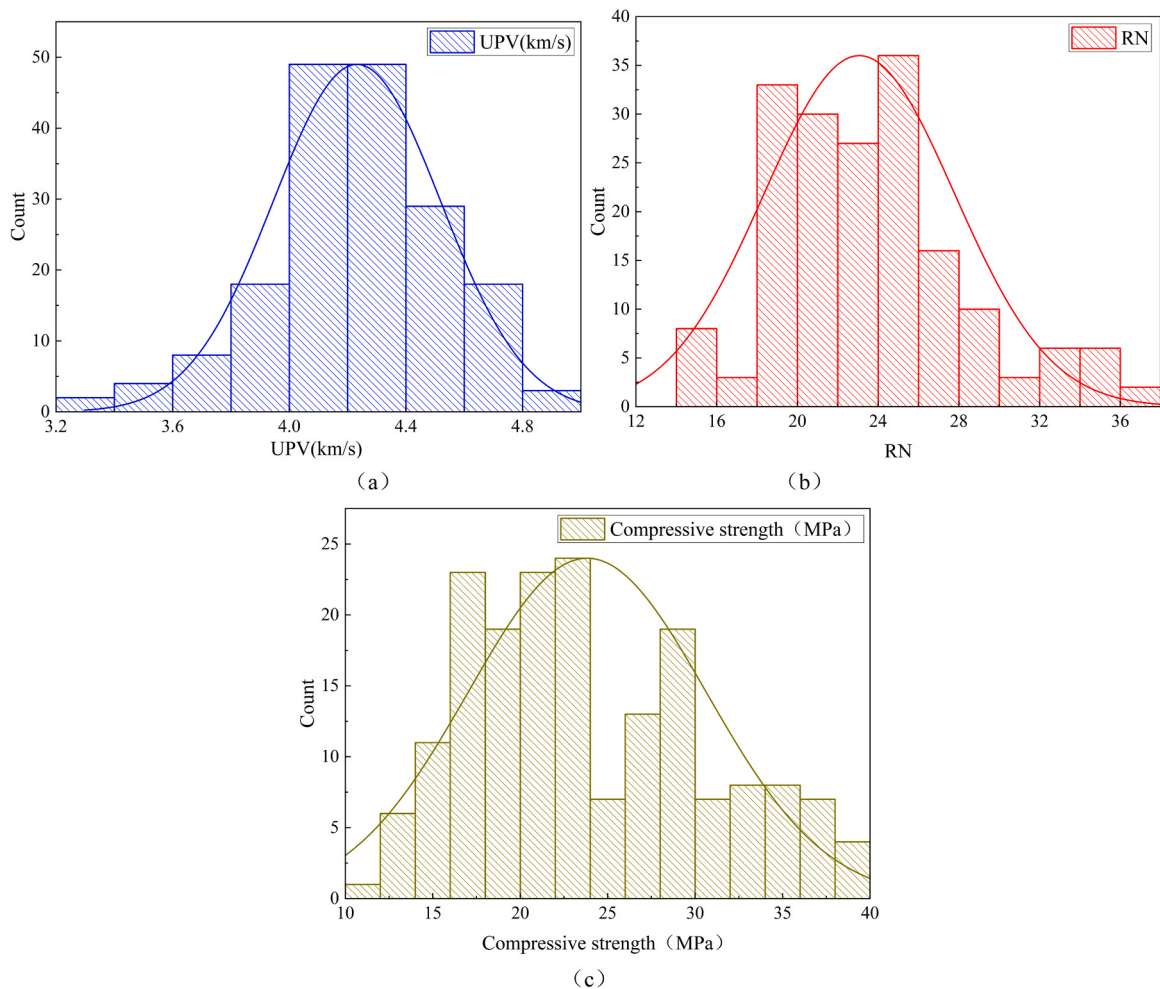


Fig. 2. Distribution of sample data: (a) UPV; (b) RN; (c) Compressive strength.

Table 3
Pearson correlation coefficient calculation results.

Variant	UPV	RN	Compressive strength (CS)
UPV	1.00	0.7745	0.7881
RN	0.7745	1.00	0.9197
Compressive strength (CS)	0.7881	0.9197	1.00

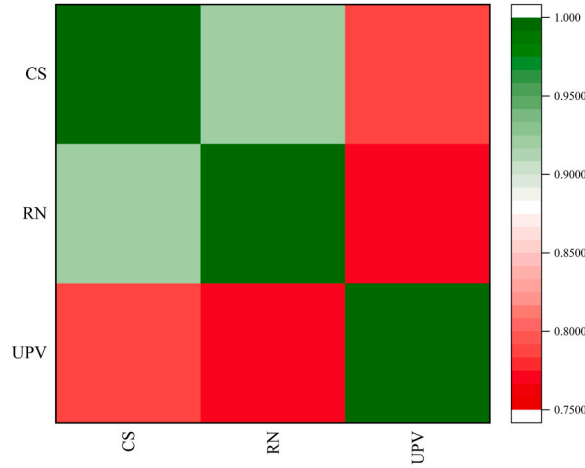


Fig. 3. Heat map of Pearson correlation coefficient of sample data.

There is a significant difference between the predicted values of compressive strength and the actual values obtained through destructive testing.

4. Methodology

To comprehensively analyze the relationship between UPV, RN, and concrete compressive strength, this study employs a machine learning approach to predict concrete compressive strength. This section focuses on the machine learning methodology used in the study and the evaluation indicators for the performance of the prediction model. The dataset $S = \{(x_i, y_i), i=1, 2, \dots, l\}$ is utilized, where x_i represents the influencing factors of the i th training sample, including concrete mix proportion design, UPV, and RN, serving as the model input. Meanwhile, y_i represents the corresponding concrete compressive strength, serving as the model output. The model sample data is divided into a training set and a test set in a 3:1 ratio. The flowchart depicting the model building process is displayed in Fig. 4.

4.1. Support vector regression

Support Vector Machine is a statistical learning theory proposed by Vapnik [58], which is based on the criterion of structural risk minimization [58,59]. This method aims to minimize both the structural risk and the sample error, thereby improving the generalization ability of the model without being limited by the dimensionality of the data. To address regression fitting problems using SVM, Vapnik et al. [58] introduced an insensitive loss function ϵ , resulting in a regression-type support vector machine. This model exhibits superior performance and effectiveness. In SVM regression fitting, the main idea is to determine an optimal plane that minimizes the error between the training samples and the optimal plane [60].

Without loss of generality, let the training set consist of l samples, denoted as $\{(x_i, y_i), i=1, 2, \dots, l\}$, where $x_i (x_i \in \mathbf{R}^d)$ is the input column vector of the i th training sample, given by $x_i = [x_i^1, x_i^2, \dots, x_i^d]^T$, and $y_i \in \mathbf{R}$ is the corresponding output value.

The objective is to build a linear regression function, $f(x)$, in a high-dimensional feature space [61,62], which can be represented as:

$$f(x) = \omega \Phi(x) + b \quad (2)$$

where $\Phi(x)$ is a nonlinear mapping function.

To define the loss function, ϵ -insensitive loss function, ϵ is introduced. Fig. 5 illustrates the ϵ -insensitive loss function set in SVR for the given data, which can be expressed as:

$$L(f(x), y, \epsilon) = \begin{cases} 0, & |y - f(x)| \leq \epsilon \\ |y - f(x)| - \epsilon, & |y - f(x)| > \epsilon \end{cases} \quad (3)$$

In Eq. (3), $f(x)$ represents the predicted value calculated by the regression function, and y represents the true value.

To solve the optimization problem, slack variables ξ_i and ξ_i^* are introduced, and the values of ω and b are computed using Eq. (4):

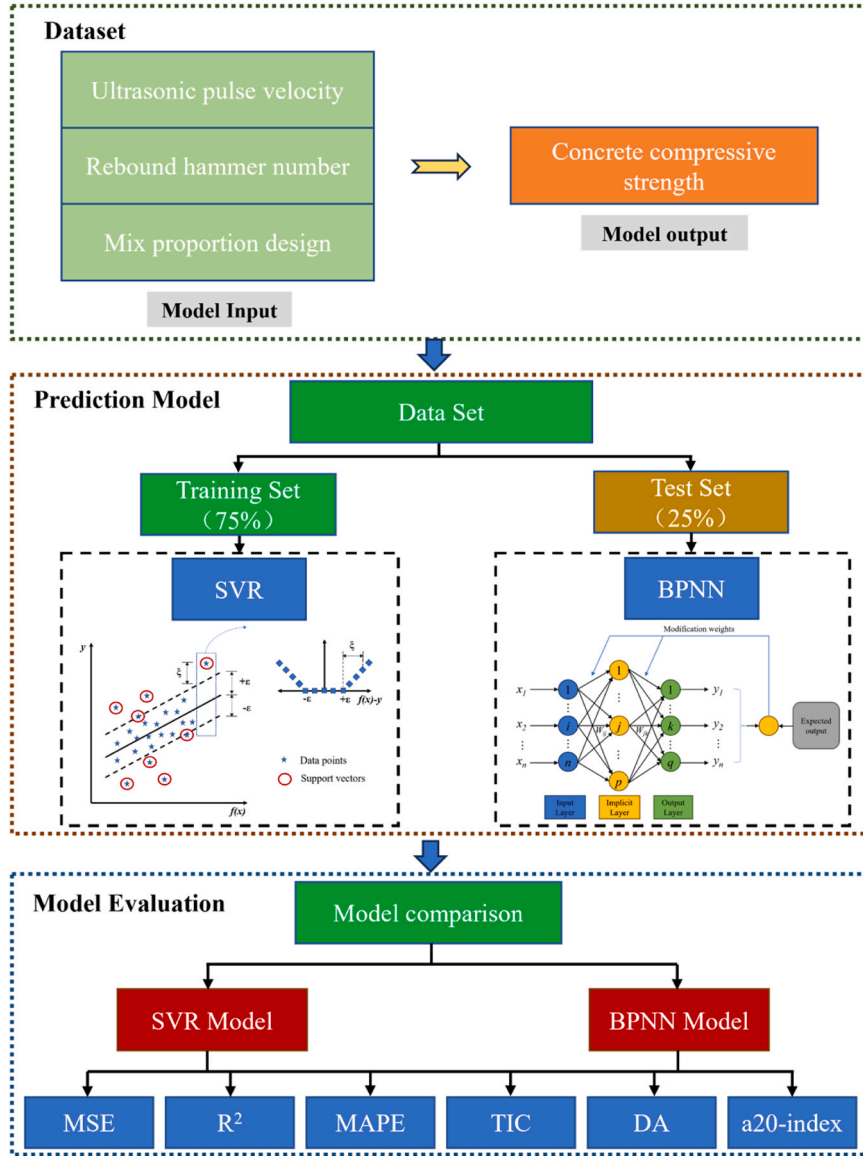
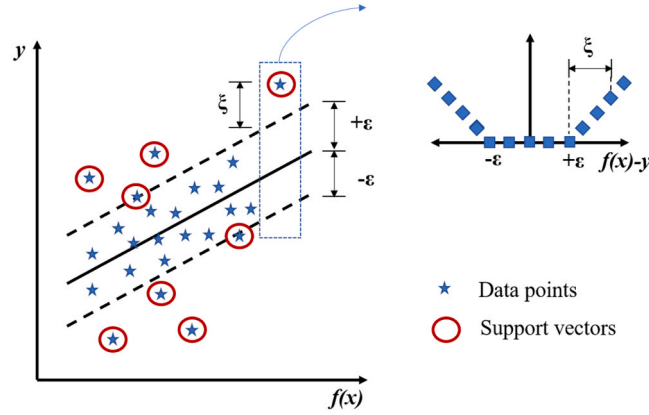


Fig. 4. The study flowchart.

$$\begin{cases} \min \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \\ s.t. \begin{cases} y_i - \omega \Phi(x_i) - b \leq \epsilon + \xi_i, & i = 1, 2, \dots, l \\ -y_i + \omega \Phi(x_i) + b \leq \epsilon + \xi_i^* \\ \xi_i \geq 0, \xi_i^* \geq 0 \end{cases} \end{cases} \quad (4)$$

where C is the penalty factor, the larger its value means the larger the penalty for the sample error over ϵ , and ϵ denotes the error of the regression function.

The Lagrange function is introduced to solve the optimization problem, converting it to a dyadic form.

Fig. 5. Insensitive loss function in SVR ε .

$$\left\{ \begin{array}{l} \max_{\alpha, \alpha^*} \left[-\frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l (\alpha_i - \alpha_i^*) (\alpha_j - \alpha_j^*) K(x_i, x_j) - \sum_{i=1}^l (\alpha_i + \alpha_i^*) \varepsilon + \sum_{i=1}^l (\alpha_i - \alpha_i^*) y_i \right] \\ \text{s.t.} \left\{ \begin{array}{l} \sum_{i=1}^l (\alpha_i - \alpha_i^*) y_i = 0 \\ 0 \leq \alpha_i \leq C \\ 0 \leq \alpha_i^* \leq C \end{array} \right. \end{array} \right. \quad (5)$$

The kernel function $K(x_i, y_j) = \Phi(x_i) \Phi(y_j)$ is utilized in SVR, with commonly used options including the polynomial kernel function, linear kernel function, Radial Basis Function (RBF), and Sigmoid kernel function [63]. Among these choices, the RBF kernel function demonstrates excellent generalization ability and yields the best performance for the model [34]. However, when using the RBF kernel function, the ill-conditioned problem is often faced due to the presence of noise or redundant features in the data. The complexity of the model is controlled by choosing an appropriate penalty factor to avoid overfitting. The criteria for selecting the penalty factor C is usually done by cross validation. When the performance of the model is the same, the parameter combination with a smaller penalty factor C is preferred.

Let the optimal solution be denoted as $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_l]$, $\alpha^* = [\alpha_1^*, \alpha_2^*, \dots, \alpha_l^*]$.

$$\omega^* = \sum_{i=1}^l (\alpha_i - \alpha_i^*) \Phi(x_i) \quad (6)$$

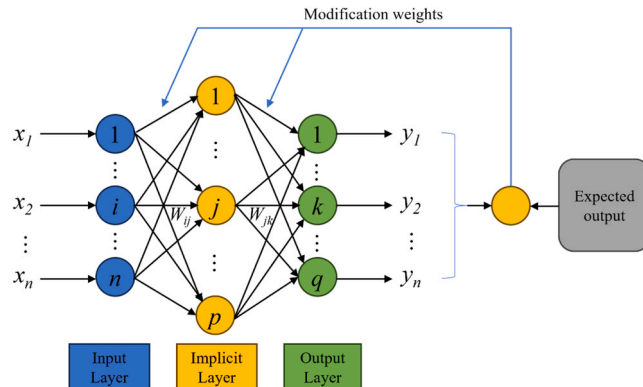


Fig. 6. Structure of BP neural network.

$$b^* = \frac{1}{N_{\text{rsv}}} \left\{ \sum_{0 < \alpha_i < C} \left[y_i - \sum_{x_j \in \text{SV}} (\alpha_i - \alpha_i^*) K(x_i, x_j) - \varepsilon \right] + \sum_{0 < \alpha_j < C} \left[y_j - \sum_{x_i \in \text{SV}} (\alpha_j - \alpha_j^*) K(x_i, x_j) + \varepsilon \right] \right\} \quad (7)$$

The number of support vectors, denoted as N_{rsv} , represents the total count of these non-zero parameters. The regression function can then be expressed as a combination of these support vectors.

$$f(x) = \omega^* \Phi(x) + b^* = \sum_{i=1}^l (\alpha_i - \alpha_i^*) \Phi(x_i) \Phi(x) + b^* = \sum_{i=1}^l (\alpha_i - \alpha_i^*) K(x_i, x) + b^* \quad (8)$$

There exists a fraction of non-zero parameters $(\alpha_i - \alpha_i^*)$ corresponding to the samples x_i as support vectors for the research problem.

4.2. BP neural network

Considering the advantages of neural networks in handling complex nonlinear relationships, this study incorporates neural network modeling into the prediction of concrete strength. The artificial neural network model is a mathematical tool that emulates the functions of the human brain, such as learning, reasoning, and performing parallel computations [5,49]. The back propagation (BP) neural network consists of an input layer, an intermediate layer (also known as the hidden layer), and an output layer, as shown in Fig. 6. In Fig. 6, x_i represents the input vector $X = (x_1, x_2, \dots, x_i, \dots, x_n)$, y_i represents the output vector $Y = (y_1, y_2, \dots, y_i, \dots, y_n)$, and W_{ij} and W_{jk} represent the connection weights between the input layer and the hidden layer. When considering concrete mix proportion design, the architecture of BPNN is divided into 3 layers with the number of neurons in each layer being 7, 10 and 1, respectively.

The learning process of the BPNN consists of two stages [31,47]. In the first stage, information is propagated forward from the input layer through the hidden layer to the output layer. By inputting the sample data, the input vector is processed through the designed network structure and the weights and thresholds are adjusted iteratively to achieve sample learning. In the second stage, information is propagated backward from the output layer to the input layer, following a specific rule to modify the initial weights and thresholds of the algorithm. This process aims to facilitate the convergence of the algorithm. The first stage of the learning process is completed when information flows from the input layer to the output layer. If there is a discrepancy between the output result of the output layer and the expected result, the error is propagated backward from the output layer to the input layer through the intermediate layer. The weights are adjusted layer by layer using the gradient descent algorithm to complete the second stage of the learning process. These two stages are repeated alternately until convergence is achieved, signaling the end of the learning process and the termination of network training.

4.3. Model evaluation

The SVR and BPNN prediction models are implemented using MATLAB. The mean square error (MSE) and coefficient of determination (R^2) are calculated for both the training and test sets. The MSE value ranges from $[0, +\infty)$, with a value of 0 indicating a perfect match between the predicted and true values. Therefore, a smaller MSE value corresponds to a smaller error. On the other hand, the R^2 measures the proportion of variation in the dependent variable that can be explained by the independent variable through the regression relationship. A higher R^2 value indicates better model performance. The formulas for MSE and R^2 are as follows [64–66]:

$$MSE = \frac{1}{l} \sum_{i=1}^l (\hat{y}_i - y_i)^2 \quad (9)$$

$$R^2 = \frac{\left(l \sum_{i=1}^l \hat{y}_i y_i - \sum_{i=1}^l \hat{y}_i \sum_{i=1}^l y_i \right)^2}{\left(l \sum_{i=1}^l \hat{y}_i^2 - \left(\sum_{i=1}^l \hat{y}_i \right)^2 \right) \left(l \sum_{i=1}^l y_i^2 - \left(\sum_{i=1}^l y_i \right)^2 \right)} \quad (10)$$

Where l is the number of samples in the test set, $y_i (i = 1, 2, \dots, l)$ is the true value of the i th sample, $\hat{y}_i (i = 1, 2, \dots, l)$ is the predicted value of the i th sample.

To further evaluate the prediction effect, other statistical indicators commonly used in machine learning are adopted. These include mean absolute percentage error (MAPE), Theil inequality coefficient (TIC), and directional accuracy (DA) [67–71].

MAPE is calculated through Eq. (11). The smaller value of MAPE indicates the smaller error, the better the regression of the model, and the higher the prediction accuracy. The advantage of MAPE is mainly that it does not change with the scaling of the dependent variable.

$$MAPE = \frac{100\%}{l} \sum_{i=1}^l \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (11)$$

TIC is calculated using Eq. (12). The value of TIC is in the range of $[0, 1]$. The smaller the value of TIC, the smaller the error between the predicted value and the real value, and the higher the prediction accuracy.

$$TIC = \frac{\sqrt{\frac{1}{l} \sum_{i=1}^l (\hat{y}_i - y_i)^2}}{\sqrt{\frac{1}{l} \sum_{i=1}^l (\hat{y}_i)^2} + \sqrt{\frac{1}{l} \sum_{i=1}^l (y_i)^2}} \quad (12)$$

DA is used to judge whether the predicted and true values change in the same direction compared to the previous period. Calculated by Eqs. (13) and (14), the closer its value is to 1, the higher the model prediction accuracy is.

$$DA = \frac{1}{l} \sum_{i=1}^l A_i \quad (13)$$

$$A_i = \begin{cases} 1 & (\hat{y}_{i+1} - y_i)(\hat{y}_{i+1} - y_i) > 0 \\ 0 & (\hat{y}_{i+1} - y_i)(\hat{y}_{i+1} - y_i) \leq 0 \end{cases} \quad (14)$$

Where $\hat{y}_i$ is the predicted value and y_i is the measured value.

In addition to the above metrics, a new model performance evaluation metric was calculated in this study. The a20-index was calculated using the following equation [72–75].

$$a20 - \text{index} = \frac{m20}{M} \quad (15)$$

where m20 is the number of samples whose ratio of true to predicted values is between 0.80 and 1.20, and M is the total number of samples in the dataset. The predictive model is a perfect model when the value of a20-index is 1. The physical engineering meaning of a20-index denotes the percentage of samples in which the error of the predicted value with respect to the true value is controlled to $\pm 20\%$.

5. Results and discussion

5.1. Model prediction results without considering the concrete mix proportion design

Combined with the established concrete UPV, RN, and compressive strength dataset $S = \{(x_i, y_i), i=1, 2, \dots, l\}$. The SVR model and BPNN model were used to predict the compressive strength of concrete at the ages of 7d, 28d, and 90d, respectively, without considering the concrete mix proportion design. Furthermore, the predicted results of concrete strength under different prediction conditions are shown in Fig. 7, Fig. 8, Fig. 9, and Fig. 10.

As can be seen from Fig. 7 and Fig. 8, the results of predicting the 7d compressive strength of concrete by comparing the SVR model and the BPNN model without considering the concrete mix proportion design. The R^2 of the training samples are 85.6 % and 85.2 %, and the prediction model cannot accurately capture the key patterns and laws in the data, and the model training results perform

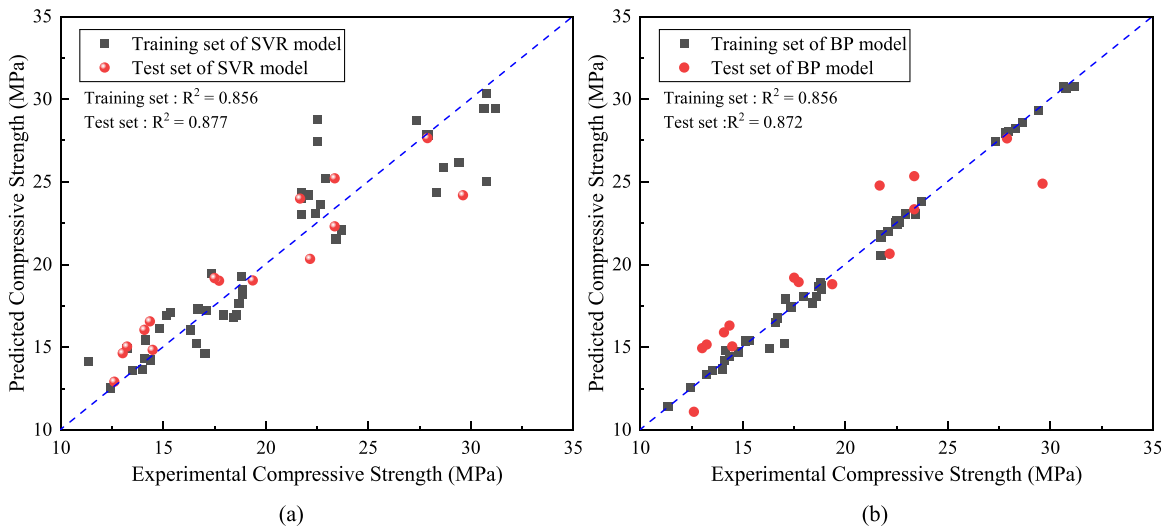


Fig. 7. Relationship between the experimental values and predicted values of 7d compressive strength: (a) SVR model; (b) BPNN model.

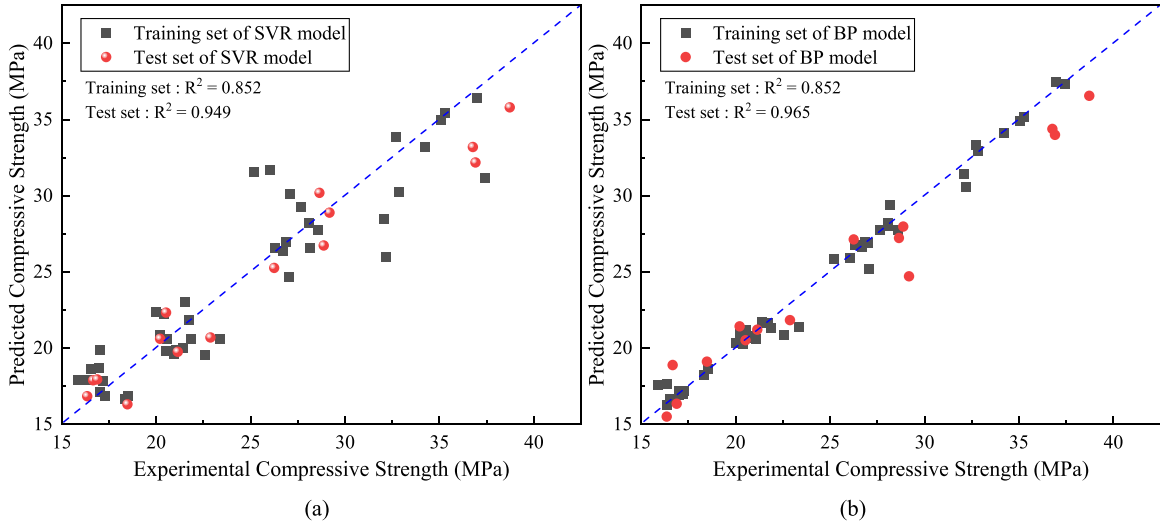


Fig. 8. Relationship between the experimental values and predicted values of 28d compressive strength: (a) SVR model; (b) BPNN model.

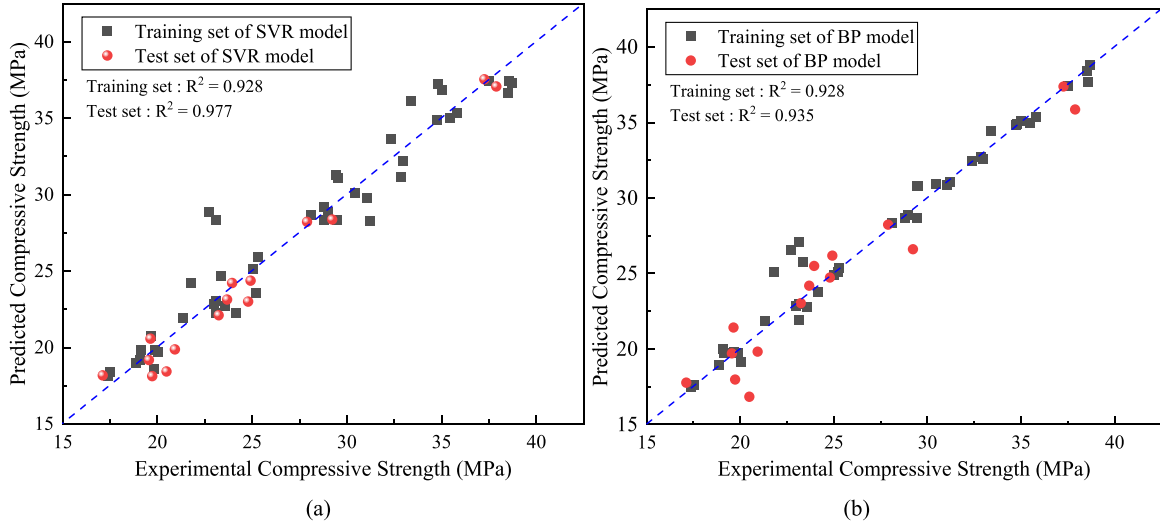


Fig. 9. Relationship between the experimental values and predicted values of 90d compressive strength: (a) SVR model; (b) BPNN model.

poorly. Furthermore, in Fig. 9, the prediction results of the two prediction models for the 90d compressive strength of concrete without considering the concrete mix proportion design are compared. The R^2 for the training samples is 92.8 %. The R^2 for the test set of the SVR prediction model is 97.7 %, and the R^2 for the training set of the corresponding BPNN model is 93.5 %. For the prediction results of concrete 90d compressive strength, the SVR model is better than the BPNN model. Similarly, in Fig. 10, the results of compressive strength prediction of concrete at all ages by the two models are compared without considering the concrete mix proportion design. The R^2 of the training samples is 90.4 %. the R^2 of the test set of the SVR prediction model is 86.0 %, and the R^2 of the training set of the corresponding BPNN model is 86.7 %. the prediction accuracies of the SVR model and the BPNN model are limited.

In summary, it can be seen that the training sets of the SVR model and the BPNN prediction model were generally fitted for the compressive strength prediction results of concrete at different curing ages without considering the concrete mix proportion design. Both models showed different degrees of overfitting and underfitting during the training process. The reason for this is mainly because the SVR model and the BPNN prediction model have a certain degree of dependence on the size of the required training dataset. When the number of model input samples is limited, the model cannot capture the key information and laws of the sample data. As a result, the sample training effect is poor, leading to poor model test performance. This requires further optimization of the control parameters and the development of relatively large training databases.

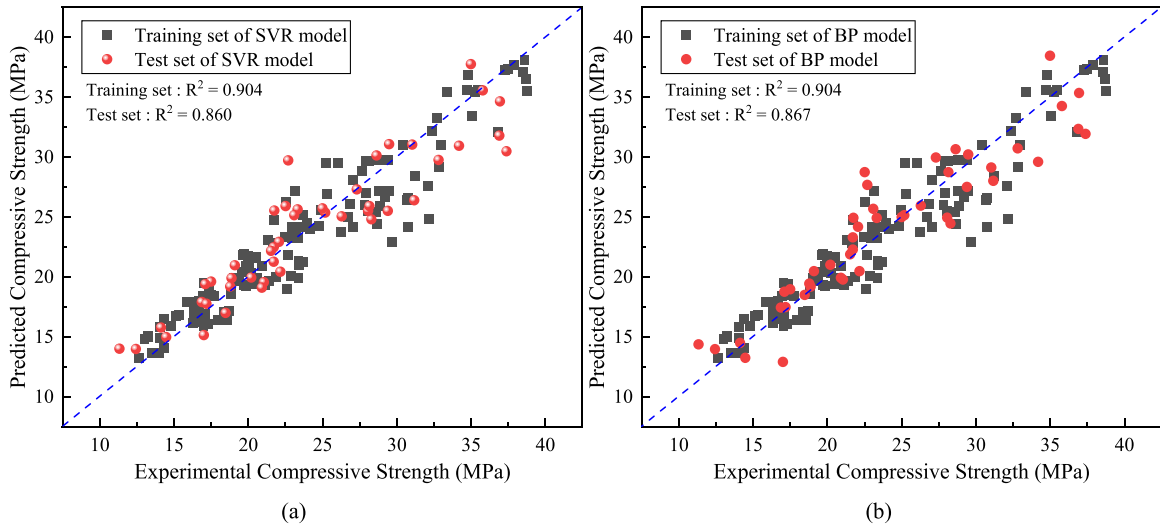


Fig. 10. Relationship between the experimental values and predicted values of all compressive strength: (a) SVR model; (b) BPNN model.

5.2. Model prediction results when considering the concrete mix proportion design

The SVR model and BPNN model were employed to predict the compressive strength of concrete at different curing ages (7d, 28d, and 90d) using the established dataset $S=\{(x_i, y_i) \mid i=1,2,\dots,l\}$, which consisted of concrete mix proportion design scheme, UPV, RN, and compressive strength. Additionally, predictions were made for overall concrete strengths without considering the effect of curing age. Furthermore, the predicted results of concrete strength by different prediction models are presented in Fig. 11, Fig. 12, Fig. 13, and Fig. 14.

In Fig. 11, a comparison is made between the results obtained from the SVR model and the BPNN model for predicting the 7d compressive strength of concrete in the context of concrete mix proportion design. It is observed that the predicted values align closely with the measured values for the training samples, resulting in an impressive R^2 value of 99.4 %. The R^2 for the SVR model's test set prediction stands at 97 %, while the BPNN model achieves an R^2 of 92.4 % on the training set. Notably, the SVR model exhibits superior performance over the BPNN model in predicting the 7d compressive strength of concrete. Moving on to Fig. 12, a comparison is made for the 28d compressive strength prediction of concrete using various models under the concrete mix proportion design scenario. The R^2 for the training samples reaches 98.9 %, while the SVR model achieves an R^2 of 96.6 % for the test set, surpassing the BPNN model's R^2 of 94.6 % on the training set for 28d compressive strength prediction. Once again, the SVR model proves to outperform the BPNN model in predicting the 28d compressive strength of concrete. Similarly, in Fig. 13, the prediction results for the 9d compressive strength of concrete are compared among different models under the concrete mix proportion design conditions. The SVR model

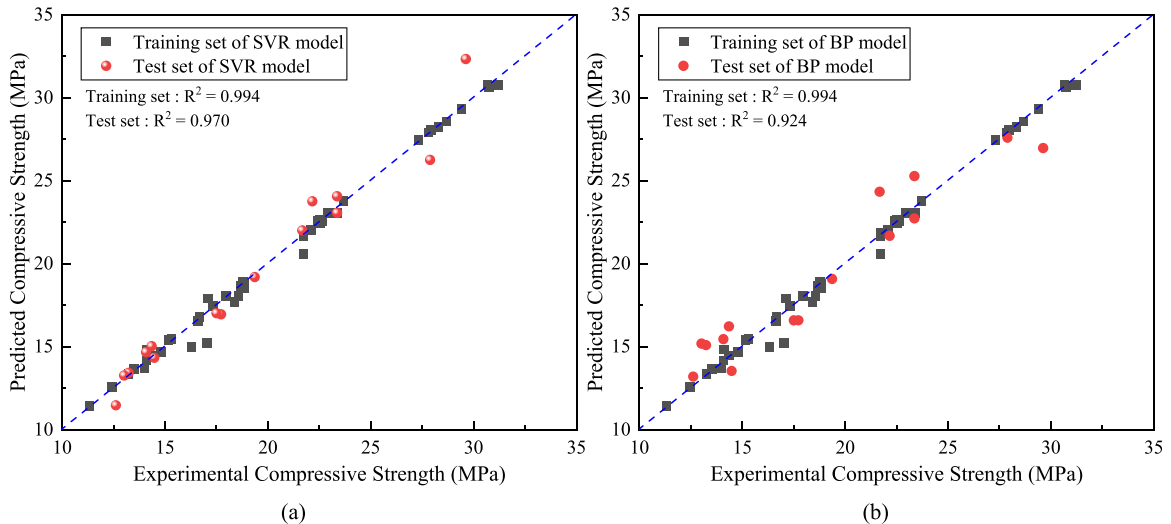


Fig. 11. Relationship between the experimental values and predicted values of 7d compressive strength: (a) SVR model; (b) BPNN model.

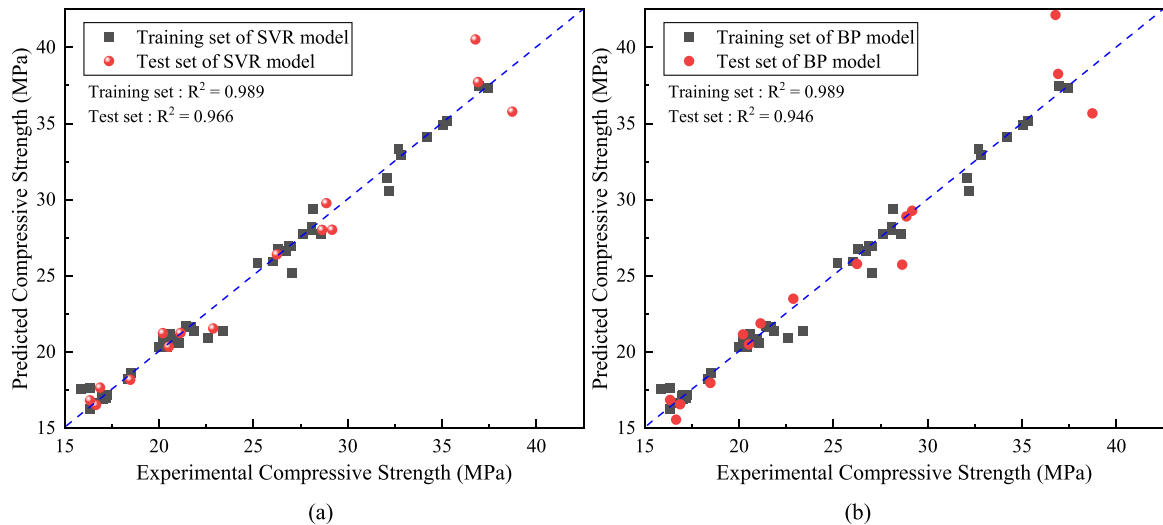


Fig. 12. Relationship between the experimental values and predicted values of 28d compressive strength: (a) SVR model; (b) BPNN model.

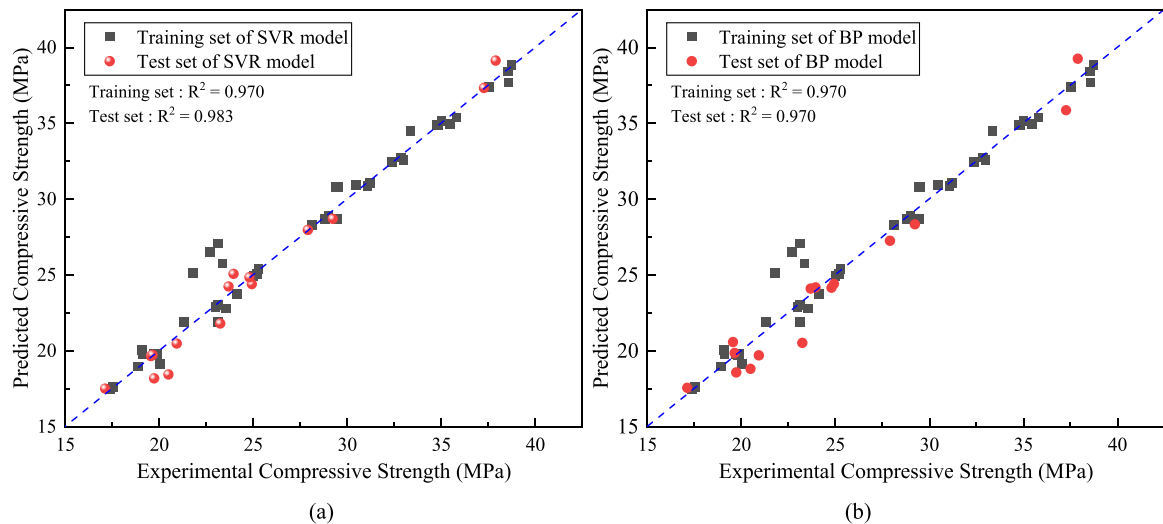


Fig. 13. Relationship between the experimental values and predicted values of 90d compressive strength: (a) SVR model; (b) BPNN model.

showcases a high R^2 of 97.0 % for the training samples and achieves an R^2 of 98.3 % for the test set, slightly exceeding the BPNN model's R^2 of 97.0 % on the training set for 90d compressive strength prediction. The SVR model demonstrates a slightly better predictive capability than the BPNN model for the 90d compressive strength of concrete. Furthermore, Fig. 14 presents a comprehensive comparison of the compressive strength prediction for concrete at all ages between the SVR model and the BPNN model under the concrete mix proportion design setting. Once again, the predicted values align closely with the measured values for the training samples, resulting in a high R^2 of 98.3 %. While the SVR model achieves an R^2 of 94.8 % for the test set, surpassing the BPNN model's R^2 of 92.4 % on the training set. It is evident that the SVR model performs better than the BPNN model in predicting the compressive strength of concrete across all ages.

In summary, the results indicate that both the SVR model and BPNN prediction model exhibit excellent fit for predicting the compressive strength of concrete at various ages under the concrete mix proportion design conditions. High consistency between predicted and measured values is observed for the training samples, with R^2 exceeding 98 %. The SVR model demonstrates superior predictive performance with R^2 values exceeding 95 % for test set samples, compared to the BPNN model's R^2 values above 92 %. Therefore, the results of the two models for concrete compressive strength prediction at different curing ages are in good agreement with the measured data. The prediction effect of the SVR model is better than that of the BPNN model. There are obvious fluctuations and data lags in the BPNN prediction model. The reason for this is mainly because the BPNN prediction model is mostly used to analyze large sample data when the number of samples tends to infinity. This requires the development of a large number of optimized control parameters and a relatively large training database. However, SVR model computational complexity depends only on the number of

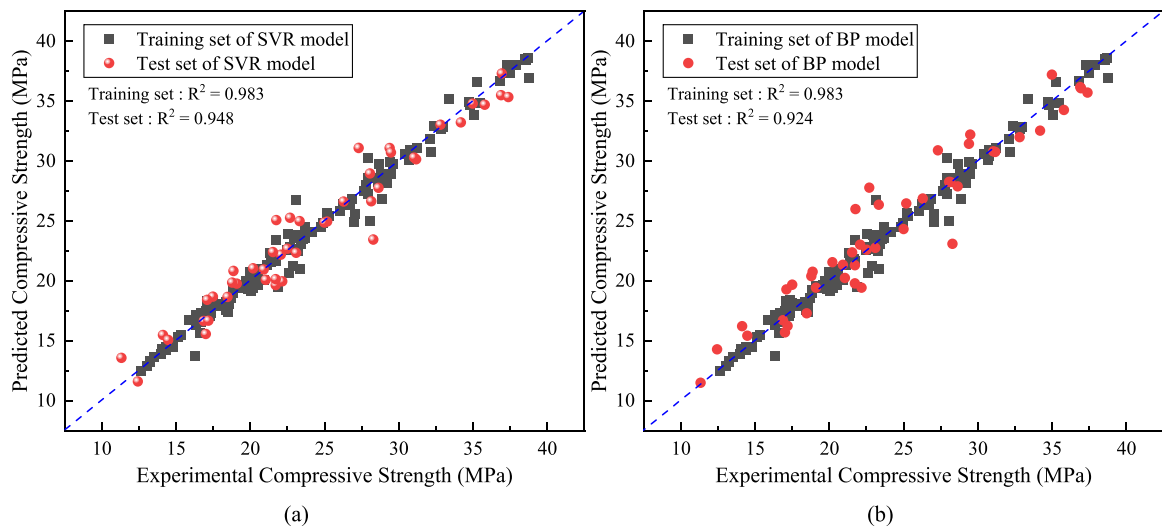


Fig. 14. Relationship between the experimental values and predicted values of all compressive strength: (a) SVR model; (b) BPNN model.

support vectors. With little dependence on the size of the required training dataset, it can effectively avoid dimensionality catastrophe, overcome the over-learning problem of artificial neural networks, and greatly simplify the regression problem, these advantages of SVR can make it completely possible to replace artificial neural networks.

To further compare the error between the prediction results of compressive strength of concrete at different curing ages and the measured values by the SVR model and BPNN model when considering the concrete mix proportion design case. Furthermore, the calculated results of the concrete strength prediction errors for different models are shown in Fig. 15 and Fig. 16.

As shown in Fig. 15, the test set sample errors were calculated by comparing the SVR model and the BPNN model for the prediction of 7d, 28d, and 90d compressive strength of concrete. It is found that the maximum value of the test set error of the SVR model is 10.11 %, and the rest of the prediction error can be controlled within 10 %. On the other hand, the maximum value of the test set error of the BPNN model is 16.34 %, and about one-sixth of the prediction errors exceed 10 %. Comparing the error calculation results of the SVR model and the BPNN model for the prediction of compressive strength of the same concrete, it is found that the test set error of the SVR model is significantly smaller than that of the BPNN model. It shows that the prediction accuracy of the SVR model is significantly higher than that of the BPNN model, and the SVR prediction model performs better.

Similarly, as depicted in Fig. 16, the results of the error calculations for the test set samples of the SVR model and the BPNN model for the prediction of the compressive strength of concrete at all curing ages are compared. It is found that the maximum value of prediction error for the test set of the SVR model is 19.4 %, and about 90 % of the test set samples have prediction error basically within 10 %. While the maximum value of the error of the test set of the BPNN model is 22.2 %, about 78 % of the samples can be controlled within 10 % of the prediction error. It shows that both the established the SVR model and BPNN model for concrete compressive strength prediction have high prediction accuracy. The prediction accuracy of the SVR model is significantly higher than that of the BPNN model, and the performance of the SVR prediction model is better. In addition, the results of the two prediction models are compared for the concrete compressive strength prediction when age is considered and when age is not considered. It is concluded that the prediction accuracy of the two prediction models is significantly higher when the effect of curing age is considered than when the effect of age is not considered. The reason for this is mainly because the discrete degree of concrete compressive strength sample data under the same curing age is smaller, and the prediction results of the model training set and test set are better.

5.3. Evaluation of prediction results

The prediction performance of the two models when considering the concrete mix proportion design case is further evaluated using commonly used evaluation metrics for machine learning. The calculated a20-index value for both the SVR model and BPNN model considering the concrete mix proportion design is 1. It shows that the error of the predicted value relative to the true value for both the models can be controlled within ± 20 %. Both prediction models are perfect models. In addition, the radar plots of model performance evaluation corresponding to MSE, R^2 , MAPE, TIC and DA are shown in Fig. 17.

As shown in Fig. 17, the evaluation metrics computed for the training and test sets of the models include MSE, R^2 , MAPE, TIC, and DA. By comparing the prediction performance evaluation indexes of the SVR model and BPNN model, it was found that for the prediction of compressive strength of concrete with different curing ages, except for the directional accuracy DA, the prediction of the SVR model was better than that of the BPNN model in all evaluation indexes. The error between the predicted and measured values of the SVR model was smaller, indicating that the SVR model has better performance and higher accuracy in predicting concrete compressive strength. Furthermore, the comparison between the SVR model and the BPNN model for predicting the compressive strength of concrete with and without considering curing age showed that the prediction results of both models were significantly

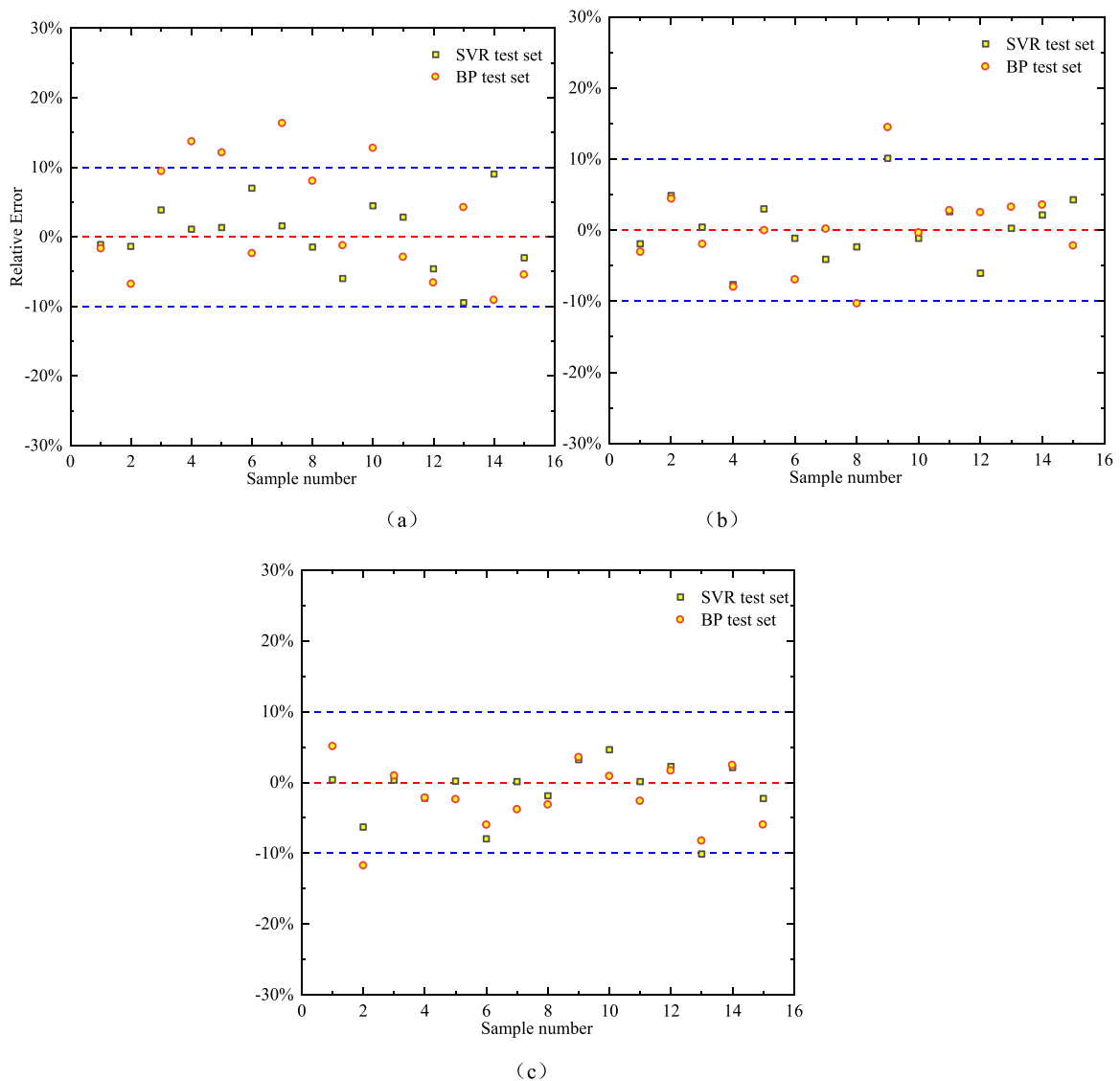


Fig. 15. Prediction errors of concrete compressive strength at different curing ages for two models: (a) 7d compressive strength; (b) 28d compressive strength; (c) 90d compressive strength.

better when considering the effect of curing age. Additionally, the SVR prediction model exhibited stronger paradigm ability compared to the BPNN prediction model, and the algorithm was robust and applicable to concrete compressive strength prediction.

6. Discussion

In Europe, America, Japan, and other developed countries, the construction of high-rise buildings and infrastructure has undergone large-scale rapid development, which has now transitioned to the stage of maintenance and repair of existing structures. The repair and renovation of existing building structures have emerged as a prominent research area in current engineering construction. The substantial economic burden brought on by the deterioration of concrete structures has far exceeded initial expectations, prompting widespread attention to the durability of concrete structures as a primary indicator of structural quality. NDT techniques are widely employed to evaluate the quality of concrete structures, particularly in terms of concrete compressive strength.

Existing studies have revealed that concrete mix design and curing age are often overlooked in predicting concrete compressive strength using NDT and machine learning methods, even though they significantly impact the strength prediction. Therefore, this study aims to introduce an effective approach for predicting concrete compressive strength by integrating concrete mix design and NDT test results. The study establishes a SVR model for concrete compressive strength and a BPNN prediction model based on NDT test outcomes, considering the influence of concrete mix design and curing age. Utilizing the established concrete mix design scheme, a dataset $S=\{(x_i, y_i), i=1,2,\dots,l\}$ comprising concrete UPV, RN, and compressive strength data is employed. Concrete compressive

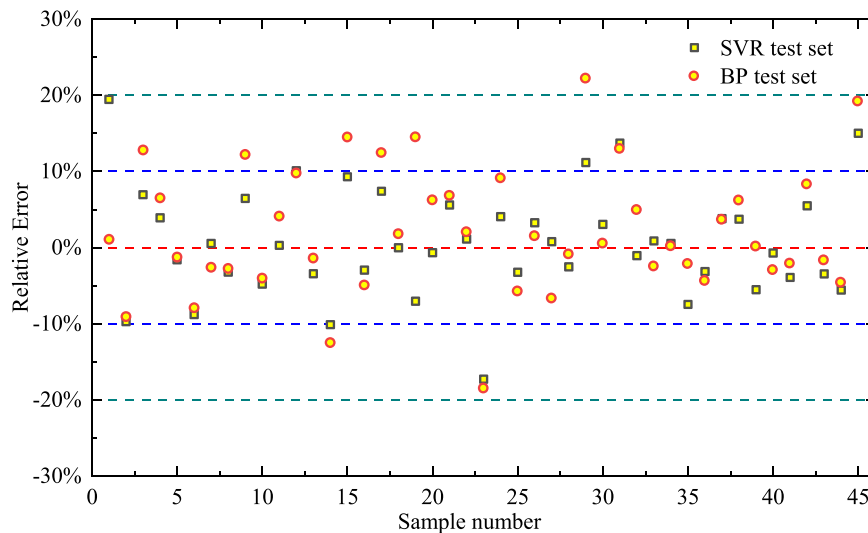


Fig. 16. Prediction errors of concrete compressive strength at all curing ages.

strength at 7d, 28d, and 90d, both with and without accounting for age effects, is predicted using the two models, with and without concrete mix proportion design considerations. MSE and R^2 values are obtained for various model training and testing sets, and the impact of concrete mix design consideration on the predictive model performance metrics is presented in Table 4.

Table 4 illustrates the comparison of compressive strength predictions by the SVR model and BPNN model for concrete, with and without consideration of concrete mix proportion design. The predictive models generally fit the training set concrete data well, yielding R^2 values ranging from 85 % to 93 % without mix proportion design consideration. However, when factoring in concrete mix proportion design, the predictive models demonstrate a superior fit to the training data, with R^2 coefficients exceeding 97 % and reaching a maximum of 99.4 %. Additionally, R^2 values for concrete compressive strength prediction without mix proportion design consideration range from 86 % to 98 %, indicating moderate performance. Notably, the SVR prediction model achieves R^2 values above 95 % for concrete compressive strength predictions when considering concrete mix proportion design, while the BPNN model achieves R^2 values between 92 % and 97 %. A comparison between the SVR and BPNN models reveals higher prediction accuracy for the SVR model in predicting concrete compressive strength, emphasizing its superior performance. Thus, the SVR prediction model outperforms the BPNN model in concrete compressive strength prediction applications.

The prediction results revealed that the two prediction models generally fit the training set for predicting concrete compressive strength without considering the concrete mix proportion design. However, both models displayed overfitting and underfitting phenomena during the training process. This is primarily attributed to the models' inability to capture key information and patterns from the limited number of input samples. Consequently, a relatively large database was developed to address this issue. Subsequently, the SVR model and the BPNN model exhibited improved fitting of the training set when considering concrete mix proportion design. Both prediction models demonstrated higher accuracy in predicting the compressive strength of concrete at all curing ages, with the SVR model significantly outperforming the BPNN model.

Comparison with the relevant literature, specifically the work of Ali Poorarbabi and Mohammadreza Ghasemi et al. [1], who aimed to estimate the compressive strength of concrete using the response surface method by testing the UPV and RN of concrete specimens, showed that the R^2 of their prediction model for concrete compressive strength at 7, 28, and 90 days were 86.3 %, 87.1 %, and 93.4 % respectively, with an overall R^2 of 86.8 %. However, their proposed concrete compressive strength prediction model exhibited significantly lower accuracy compared to the SVR and BPNN prediction models proposed in our study. Asteris et al. [5] developed six soft computing models to predict concrete compressive strength using ultrasonic pulse velocity and rebound hammer tests. The most accurate prediction of concrete compressive strength was obtained for BPNN. The corresponding coefficient of determination R^2 was 95.03 %. It was slightly lower than the coefficient of determination of 98.3 % for the SVR prediction model used in this study.

The results depict the high prediction accuracy of the concrete compressive strength prediction model established in this study, which leverages NDT of concrete and machine learning methods, and its potential for generalization. However, the limited size of the sample dataset for concrete mix proportion design, UPV, RN, and compressive strength is a notable limitation. Future research could address this by collecting a larger sample size. Additionally, with the rapid development of group intelligence methods, novel heuristic algorithms such as the Whale Optimization Algorithm and Gray Wolf Optimization Algorithm could be utilized to optimize the SVR model parameters in upcoming studies.

7. Conclusion

In many industrialized countries, the rehabilitation and retrofitting of existing building structures has become a focal point in current engineering and construction research. The evaluation of concrete compressive strength is a critical aspect of assessing the

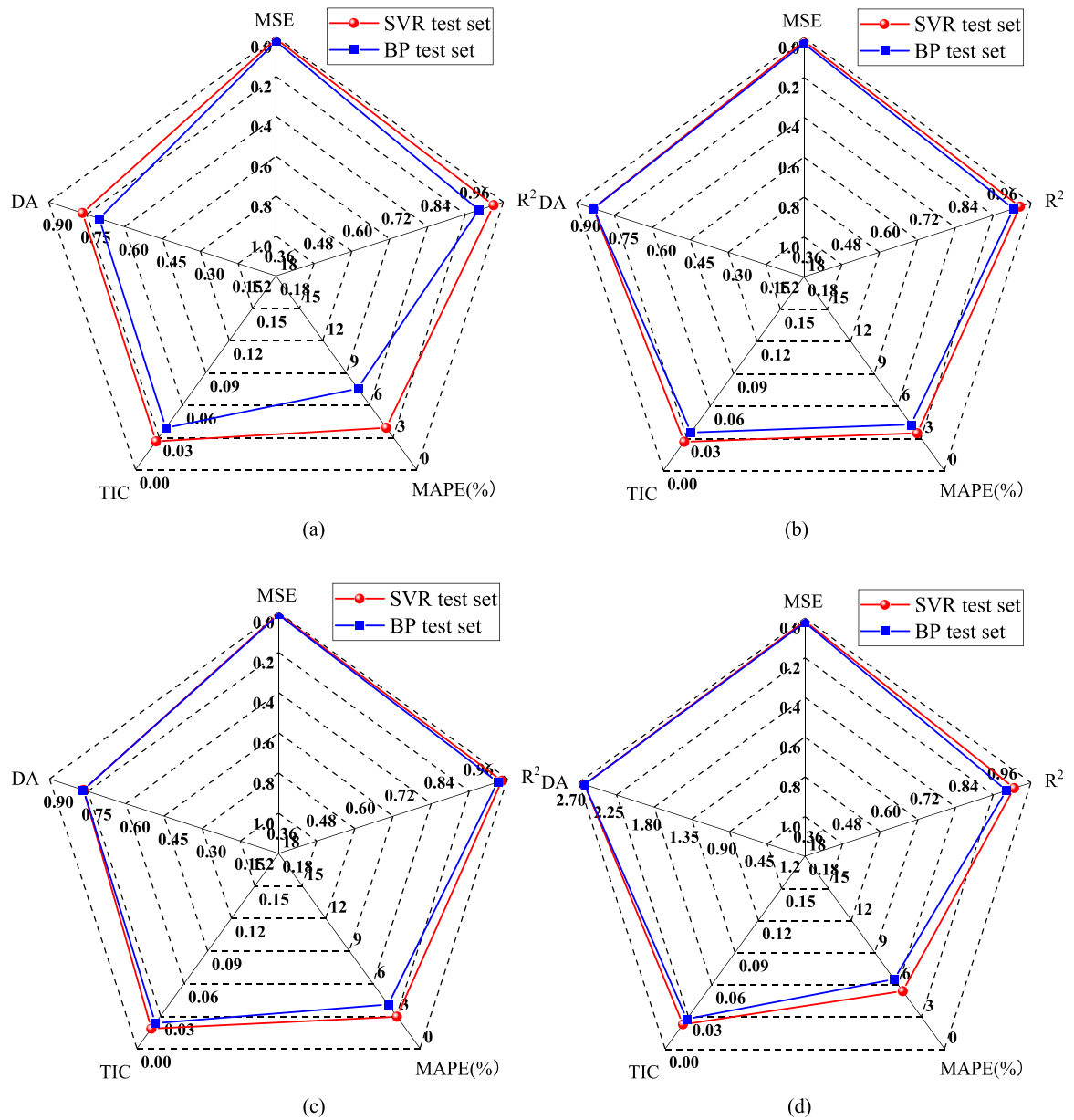


Fig. 17. Radar chart for performance assessment of concrete compressive strength prediction model: (a) 7d compressive strength; (b) 28d compressive strength; (c) 90d compressive strength; (d) compressive strength at all curing ages.

Table 4
Results of the effect of concrete mix proportion design on model performance indicators.

Sample set	Index	Without considering the concrete mix proportion design			Considering the concrete mix proportion design		
		Training set	SVR test set	BPNN test set	Training set	SVR test set	BPNN test set
7d compressive strength	MSE	0.047	0.042	0.040	0.002	0.011	0.024
	R ²	0.856	0.877	0.872	0.994	0.970	0.924
28d compressive strength	MSE	0.050	0.041	0.030	0.005	0.017	0.030
	R ²	0.852	0.949	0.965	0.989	0.966	0.946
90d compressive strength	MSE	0.026	0.010	0.022	0.012	0.008	0.012
	R ²	0.928	0.977	0.935	0.970	0.983	0.970
compressive strength at all ages	MSE	0.027	0.040	0.037	0.005	0.014	0.022
	R ²	0.904	0.860	0.867	0.983	0.948	0.924

performance of these structures. NDT is commonly used to assess the compressive strength of concrete structures. In this study, two machine learning models, SVR and BPNN, were established to predict the compressive strength of concrete by integrating NDT test results of concrete with different mix proportion designs and curing ages. The performance of the established models was further evaluated using various machine learning evaluation metrics. The following conclusions are mainly drawn:

1. A total of 180 datasets of concrete specimens with different mix proportion designs and curing ages were collected from the literature to establish the corresponding sample datasets for prediction modeling. The model inputs included concrete mix proportion design, UPV, and RN, with the concrete compressive strength utilized as the model output.
2. The established SVR model and BPNN model were employed to predict the concrete compressive strength using 75 % of the sample data as the training set and 25 % as the test set. The test set R^2 of the SVR model and the BPNN prediction model for the compressive strength of concrete at all ages was 86.0 % and 86.7 % without considering the concrete mix proportion design.
3. The fitting effect of the SVR model and BPNN model on the training set for predicting concrete compressive strength improved when considering concrete mix proportion designs. The prediction results revealed the R^2 higher than 95 % for the SVR model, while the corresponding R^2 for the BPNN prediction model ranged between 92 % and 97 %. The predicted concrete compressive strengths from both models closely aligned with the actual values, with the SVR model displaying superior prediction accuracy.
4. The prediction effects of the SVR and BPNN models were further evaluated using MAPE, TIC, DA, and a20-index. The results demonstrated that all the evaluation indexes of the SVR model were better than the BPNN model, indicating that the prediction accuracy of the SVR concrete compressive strength prediction model was higher.

The study's results indicate that the established SVR model for predicting concrete compressive strength exhibits significantly higher prediction accuracy when considering concrete mix proportion designs, and displays potential for generalization. However, the study is limited by the size of the sample dataset, and future research efforts should aim to collect more data and establish a larger model dataset.

CRedit authorship contribution statement

Kunpeng Cao: Writing – review & editing, Visualization, Validation. **Dunwen Liu:** Resources, Project administration, Funding acquisition, Conceptualization. **Wanmao Zhang:** Writing – original draft, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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